

Design Moderators of Perceived Residential Crowding and Chronic Physiological Stress Among Children

Environment and Behavior

2019, Vol. 51(5) 590–621

© The Author(s) 2019

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/0013916518824631

journals.sagepub.com/home/eab

Kimberly A. Rollings¹  and Gary W. Evans²

Abstract

Chronic crowding within housing adversely affects psychological well-being, yet little is known about how design attributes contribute to these effects, especially among children. This cross-sectional study first examined associations between residential interior density and children's ($M = 9$ years of age) perceived bedroom and home crowding. Second, analyses investigated whether interior design attributes (residential floor plan arrangement measured by space syntax [depth and permeability]; bedroom ceiling height, volume, and window area) buffer negative effects of perceived crowding on multimethodological indices of child development, including psychological distress, learned helplessness, and physiological stress. After adjusting for home type, clutter, income, gender, and age, interior density was significantly associated with perceived home and bedroom crowding. Regression results suggested that bedroom ceiling height was associated with reduced negative effects of home, but not bedroom, perceived crowding on blood pressure, epinephrine, norepinephrine, and allostatic load among participants who reported higher levels of perceived crowding.

¹University of Notre Dame, IN, USA

²Cornell University, Ithaca, NY, USA

Corresponding Author:

Kimberly A. Rollings, School of Architecture, University of Notre Dame, 114 Walsh Family Hall, Notre Dame, IN 46556, USA.

Email: krolling@nd.edu

Keywords

crowding, interior design, children, psychological health, physiological stress, space syntax

Introduction

Research demonstrates that chronic crowding experienced within a high interior density residence negatively affects psychological health and physiological stress (see reviews: Evans, 2003a, 2006; Leventhal & Newman, 2010). Young children, especially those not in day care, spend more time at home than in any other environment (Hofferth & Sandberg, 2004), yet few studies have examined whether design attributes within a home buffer negative effects of crowding on children (Evans, Lepore, & Schroeder, 1996; Schiffenbauer, Brown, Perry, Shulack, & Zanzola, 1977). Considering increasing rates of high-density living and emerging “tiny house” options, understanding how interior design attributes affect the crowding–health relation is critical to informing housing policy, especially for affordable housing. This cross-sectional study first examined associations between residential interior density (people per room) and children’s perceived crowding. We then examined whether interior design elements such as floor plan arrangement and bedroom ceiling height, volume, and window area moderated the effects of perceived crowding on multimethodological indices of child development, including psychological health and physiological stress.

Residential Interior Density and Crowding Defined

Density has been defined in a number of ways and at multiple scales, from rooms and buildings to neighborhoods and cities. In the planning field, the term “density” refers to neighborhoods and cities (i.e., number of people per acre), while “crowding” refers to the density of interior spaces (Forsyth, Oakes, Schmitz, & Hearst, 2007). In this article, however, the term *residential interior density* indicates the number of persons per room within a home. Crowding (noted as perceived crowding in the remainder of this article), therefore, is defined as a psychological response to residential interior density based on perceptions of spatial restriction due to too little space or too many people present in a space (Aiello, Epstein, & Karlin, 1974; Evans, Saegert, & Harris, 2001; Stokols, 1972a; Sundstrom, 1975). The number of residents per room, rather than square footage per person, is a measure of density that is more robustly associated with behavioral outcomes (Baum & Paulus, 1987; Evans, 2006). Among the psychosocial processes leading from high density to perceived crowding are diminished feelings of control, especially over desired

social interaction (Altman, 1975); overarousal or overstimulation (Evans, 1979); interference with goals and activities (Lepore, Evans, & Palsane, 1991; Olson, 1975; Schiffenbauer et al., 1977); or interference with socially supportive relationships among home residents, leading to social withdrawal (Evans, 2003a). The following sections discuss (a) the effects of residential interior density and crowding on children's psychological health and physiological stress and (b) previous research on design characteristics (e.g., floor plan configuration) that may moderate the impacts of high interior density and perceived crowding on human health and behavior.

Adverse Effects of Crowding, Measured by Interior Density

Many cross-sectional studies (Baum, Gatchel, Aiello, & Thompson, 1981; Baum & Valins, 1979; Evans, 2003a; Evans & Lepore, 1993; Evans et al., 1996; Evans, Palsane, Lepore, & Martin, 1989; Evans, Rhee, Forbes, Allen, & Lepore, 2000; Evans et al., 2001; Lepore et al., 1991) and two prospective, longitudinal studies (Lepore et al., 1991; Wells & Harris, 2007) found adverse effects of chronic high interior density—an indicator of crowding—on adults' psychological health. Although a substantial portion of the crowding literature focuses on adults, several studies (see reviews: Evans, 1979, 2006; Leventhal & Newman, 2010) examined interior density and children's academic performance, cognitive processes, psychological distress, helplessness, and chronic stress. Earlier work on interior density and children controlled for socioeconomic status (SES), but did not use standardized instruments to assess children's psychological health. Despite this limitation, results indicated that children from higher interior density homes had more behavioral problems in the classroom (Saegert, 1982). Greater conflict also occurred among parents and children in higher interior density homes (Booth & Edwards, 1976; Evans, Lepore, Sejwal, & Palsane, 1998; Saegert, 1982). High interior density further affects children's psychological health and development indirectly via effects on parents' behaviors. Parents in higher interior density homes tend to be more critical and less responsive to their young children (Bradley & Caldwell, 1984; Evans, Maxwell, & Hart, 1999; Wachs, 1989).

One study that did use a standardized index of psychological health found significant associations between interior density and psychological health among low-income children (Evans et al., 2001). Poor housing quality, including high interior density, is a major source of stress and worry, especially for low-income families (Evans, 2004; Evans, Lercher, & Kofler, 2002). Children living in higher interior density conditions are more susceptible to learned helplessness, likely because of greater difficulty in regulating social interactions (Evans et al., 1998; Evans et al., 2002; Evans et al., 2001; Rodin, 1976).

The relation between interior density and associated negative outcomes may vary by gender, although the direction and presence of gender effects are inconsistent. In the Evans and colleagues (2001) study, child participants from higher interior density, multiple family dwellings were less likely to persist on problem-solving tasks, but rural boys were more vulnerable to the adverse psychological outcomes of high interior density than were girls. Another study found that the association between interior density and learned helplessness was significant only for 10- to 12-year-old girls (Evans et al., 1998).

Only a few studies have explored the effects of crowding on children's physiological stress outcomes (see review: Evans, 2006). A laboratory study found that boys, but not girls, in high interior density conditions had higher skin conductance levels; levels were even more elevated with longer exposure (Aiello, Nicosia, & Thompson, 1979). Skin conductance, or electrodermal response, measures the skin's response to physiological arousal from internal or external stimuli. Another study found that 8- to 10-year-old boys and girls from higher interior density apartments, especially with greater family turmoil, had elevated overnight epinephrine and norepinephrine levels (Evans & Saegert, 2000). In an additional study of 10- to 12-year-olds, gender moderated the interior density–blood pressure relation such that increases in interior density were significantly associated with increases in blood pressure among boys when controlling for household income (Evans et al., 1998).

Several interior density studies focused on children examined gender as a moderator, controlled for SES, and utilized standardized instruments to assess children's psychological health; not all studies, however, included a wide range of residential interior densities (Evans, 2003a) or examined potential moderators that might buffer the adverse effects of interior density on children. Some evidence suggests that, conceivably, physical design elements within a residence may moderate negative effects of interior density and perceived crowding. This evidence is discussed in the following sections.

Interior Density, Perceived Crowding, and Residential Design Elements: Potential Moderators

Policies focused explicitly on limiting interior density to avoid crowded living conditions neglect potential design moderators of the density–crowding relation. Schiffenbauer and colleagues (1977) concluded that “significant variation in [perceived] crowdedness ratings with no variation in physical (interior) density indicated that (interior) density is neither a necessary nor sufficient condition for [perceived] crowding to occur” (p. 13). Schiffenbauer's team (1977) studied dormitory rooms of equal size and comparable number of people per room, noting that perceptions of crowding differed based on

usable space, greater window vistas afforded by higher floor level, and amount of natural light. Reduced usable space increased perceptions of crowding, possibly because it elicited a lack of control over room layouts that may have afforded additional privacy. Window views from higher floor levels and more natural light, conversely, reduced perceptions of crowding. Although this study suggested that residential design attributes may moderate the interior density–perceived crowding relation, results were based on self-reports from all female college students and did not consider alternative explanations for differences in crowding perceptions. These findings were replicated and expanded upon by Kaya and Erkip (2001), who found that both male and female dormitory residents perceived rooms of equal size and interior density on higher floor levels as less crowded than those on lower floor levels. Evans et al. (2002) examined the effects of interior density and dwelling type on self-reports and teacher reports of psychological distress. They found that children living in multiple family dwellings with higher interior density, independently of SES, suffered greater psychological stress compared with those in lower interior density dwellings, higher interior density single-family homes, and higher interior density row houses. In addition to physical size, usable space, floor level, and natural light, additional research offers evidence that design elements such as residential floor plan arrangement, window area, and ceiling height could buffer the adverse effects of interior density on children.

Floor plan arrangement and space syntax. Human needs to regulate social interaction suggest that, given the same square footage, reactions to interior density might vary based on floor plan arrangement (Aiello et al., 1974; Altman, 1975; Evans et al., 1996; Stokols, 1972b; Sundstrom, 1975). Several studies have linked floor plan arrangement to reduced perceptions of crowding among college students. A dormitory containing long corridors that was converted into suite arrangements resulted in reduced negative outcomes of perceived crowding, more social engagement, and less social withdrawal (Baum & Davis, 1980). Other studies found that prison inmates experienced lower levels of reported perceived crowding and less physiological stress in cubicles compared with those housed in open floor plan arrangements, both with a similar amount of square footage. This appeared to have occurred because of enhanced opportunities to regulate social interaction in the cubicles (Cox, Paulus, McCain, & Karlovac, 1982; Schaeffer, Baum, Paulus, & Gaes, 1988).

Variations in floor plan configurations in the above illustrative studies can be more formally quantified by space syntax. Space syntax describes how attributes of residential floor plan layouts, such as depth and permeability,

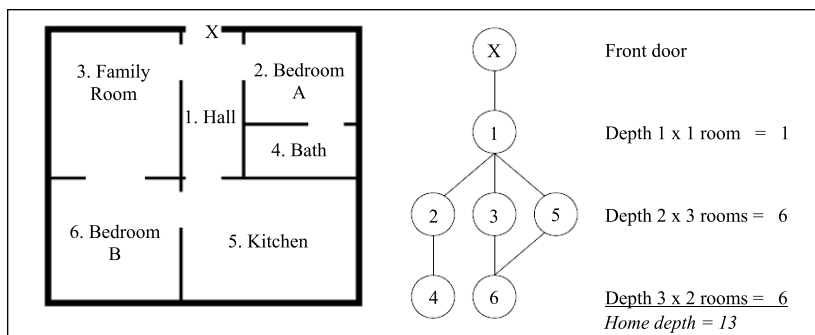


Figure 1. Example floor plan and home depth diagrams.

Note. The third row of the depth diagram, for example, indicates three rooms with a depth of 2 as one first enters the hall and then Room 2, 3, or 5.

can influence social interaction (Hillier & Hanson, 1984). Depth is defined as the number of spaces or thresholds that must be traversed before reaching a specific destination within a floor plan. Increased depth could afford desired levels of privacy and social interaction, reducing social withdrawal (Evans et al., 2000; Lepore, Merritt, Kawasaki, & Mancuso, 1990; Wells & Harris, 2007). While depth is a measure that describes floor plan layout, permeability relates more to connectivity and openness. Permeability also describes how floor plan layouts affect movement throughout a home (Hillier & Hanson, 1984). Permeability captures the accessibility of spaces and floor plan configurations, or how public and private they are, and can be quantified by the space syntax concepts of relative asymmetry (RA) and real relative asymmetry (RRA; Hanson, 2003). RA is a function of the plan's mean depth and number of rooms, and RRA is a function of RA and depth of interconnected spaces. Low values indicate highly public, integrated, and accessible spaces and plan configurations, whereas high values indicate more private spaces and plan configurations. Permeability also affords or inhibits control of stimulation and social interaction that can affect perceptions of crowding (Hillier & Hanson, 1984). An example floor plan and associated depth diagram are presented in Figure 1.

In a promising study on residential environments concerning floor plan arrangement and interior density, space syntax theory's concept of architectural depth was explored as a moderator of the interior density-psychological distress relation in college students (Evans et al., 1996). The study found that greater architectural depth moderated the interior density-psychological distress relation through added opportunities to regulate social interaction and

reduced social withdrawal (mediated moderator); therefore, negative effects of high interior density on psychological distress were reduced. This density by depth moderation matches findings that children living in higher interior density homes who can retreat to a space of their own suffer fewer negative effects of interior density (Wachs & Gruen, 1982).

Child's bedroom ceiling height and window area. Although space syntax provides useful methods to quantify differences in two-dimensional floor plan arrangements, the theory does not capture three-dimensional design elements, such as ceiling height or window area, that may affect perceptions of room size and thus potentially moderate the impacts of interior density on both perceived crowding and affective and physiological reactions to perceived crowding.

Ceiling height and volume. Adults perceive rooms with lower ceilings as smaller, offering less physical space, and therefore require more personal space; however, a room of the same size with a higher ceiling is perceived as larger and offering more personal space, so adults require less personal space and are willing to tolerate more social interaction (Savinar, 1975). Although no prior research examining effects of volume on perception of personal space or crowding was identified, volume was included in this study's analyses. Volume relates to proportions of interior space and is relevant to interior design and architectural practice.

Windows. Evidence from a few studies has suggested that natural light and views provided by windows, affected by window size, can influence perceptions of both room size and crowding (Butler & Steuerwald, 1991; Ne'eman & Hopkinson, 1970; Schiffenbauer et al., 1977). Any window view is preferred over no view (Collins, 1975), and expanded window views afforded by higher floor level were found to reduce perceptions of crowding (Schiffenbauer et al., 1977) and result in higher restorative quality ratings (Masoudinejad & Hartig, 2018). College students perceived dormitory rooms that received more natural light as less crowded than darker rooms (Baum & Davis, 1976; Schiffenbauer et al., 1977). Dormitory studies, however, did not account for window size. Other studies have also examined the effect of room size, natural light, and window view on window size preferences. In smaller rooms, a larger window in proportion to the wall was preferred (Butler & Steuerwald, 1991). Considering the lack of literature examining connections between interior density and children's perceived crowding, as well as the potential for design attributes to buffer negative effects of children's perceived crowding on indices of child development, the following exploratory study was conducted.

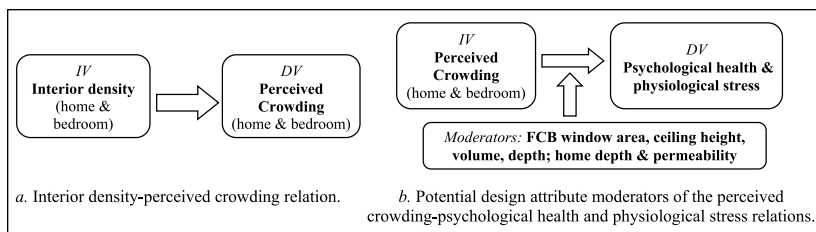


Figure 2. Study hypotheses diagrams.

Note. IV = independent variable; DV = dependent variable; FCB = focus child's bedroom.

Current Study: Research Aims

The present study had two aims. Aim 1 examined whether residential interior density was, in fact, a significant indicator of children's perceived home and bedroom crowding among the study sample. Although interior density is often used as a proxy for perceived crowding (Baum & Paulus, 1987; Evans et al., 2001), prior work indicates that interior density may not be necessary or sufficient for perceived crowding to occur (Schiffenbauer et al., 1977; Stokols, 1972b). Few studies have empirically tested associations between residential interior density and perceived crowding among children. Whether and to what degree children perceive crowding and are affected by those perceptions are understudied. Initial analyses explored the hypothesis that higher residential interior densities are strongly correlated with and predict children's perceived home and bedroom crowding (Figure 2a). If the density–perceived crowding association is weak among children, policies attempting to minimize negative effects of perceived crowding may need to address factors in addition to residential interior density, such as interior design quality.

Aim 2 then explored the potential of design elements to moderate children's psychological health and chronic physiological stress outcomes related to perceived crowding. Research about how residential design attributes buffer adverse effects of perceived crowding among children is sparse. Conceivably, design attributes that may affect social interaction within a home, as well as perceptions of space, could moderate children's responses to high residential interior density. Exploratory Aim 2 analyses assessed whether the perceived crowding–psychological health and perceived crowding–physiological stress relations among children were moderated by floor plan arrangement (depth and permeability) and the focus child's bedroom (FCB) window area, ceiling height (lowest point), and volume (Figure 2b). The hypothesis was that greater FCB depth, window area, ceiling height, and volume would moderate negative effects of perceived FCB crowding, and that

mean home depth and permeability would buffer the negative effects of perceived home crowding.

Method

Research Design and Participants

Secondary, cross-sectional data were extracted from the first wave of a longitudinal data set (Wave 1, 1995-1999; mean age = 9 years). Home interviews with 341 low- and middle-income children and their mothers were originally conducted during two home visits by trained research assistants following a uniform protocol (Evans, 2003b). Data were gathered on housing quality, psychological health, physiological stress, and demographics of one "focus child" and the child's mother. The majority of participants (88%) were White and resided in primarily rural areas throughout upstate New York. Fifty percent of participants resided in detached, single-family homes; others resided in duplexes, trailers, or multifamily residences (e.g., apartment buildings between one and four stories). Trained research assistants also sketched participants' home floor plans and recorded information about children's bedroom design for a subset of 181 children. Floor plans, bedroom design information, and outcome data for this subset of 181 children were analyzed in the current study.

Constructs and Measures

Residential interior density is defined, for the purposes of this study, as the number of persons per room within a home. Perceived crowding is defined as an individual's perceptions of spatial restriction due to too little space or too many people present in a space. The nine-item, 3-point perceived crowding scale used in this study (Online Appendix A) was modeled on a reliable and valid index of adults' perceived crowding, developed by Evans and colleagues (2000), that is sensitive to both field and laboratory levels of interior density. Perceived crowding scale items were developed via an iterative process with a separate sample of children—of similar age to children in this study's sample—that included ascertaining whether or not children understood each scale item. The process included repeating children's answers back to them and then asking what they meant when they provided their answer. A subset of items with high levels of inconsistency between the rating and the explanation were dropped from the scale. The scale has adequate internal consistency ($\alpha = .72$). Scale validity is indicated by its correlation with residential density ($r = -.35$) and the fact that the perceived crowding

items load on a separate principal components factor when run in a cluster with items measuring stress and psychological health.

Additional predictors, covariates, and outcome variables are described in Table 1. Home type (four categories) and clutter were included in regression models to account for potential design differences between home types, such as amount of available space, number and types of available spaces, noise from adjacent units, ceiling height, and clutter, that could conceivably affect a child's perception of crowding and the perceived crowding–design attribute interactions. Income-to-needs ratio, age, and gender were also included as demographic covariates. Results were adjusted for SES as in previous literature. Because the age of participants ranged from 6 to 12 years in this study, age was also included in regression models to account for differences in body size, developmental stage, and potentially higher rates of room sharing among younger versus older children. In addition, gender was included as a covariate for three reasons. Although previous literature documents an inconsistent pattern of how the effects of perceived crowding and high interior density might vary by gender, responses to high interior density could vary based on differences in gender roles and expectations within the home. Boys who spend more time outdoors than girls may also be affected by interior density differently (Cleland et al., 2010). Furthermore, physiological outcomes can vary by gender (e.g., body mass index [BMI] used to calculate allostatic load). The following sections describe the dependent variables that measured psychological health and chronic physiological stress.

Psychological health outcomes. Children's psychological health was operationalized using three age-appropriate measures gathered by trained researchers during "Wave 1" of the larger longitudinal study. First, the Rutter Children's Behavior Questionnaire (Cronbach's $\alpha = .83$; Boyle & Jones, 1985; Rutter, Tizard, & Whitmore, 1970), a widely used standardized instrument with well-documented psychometric properties, measures psychological health in nonclinical samples of young children (Boyle & Jones, 1985; Evans et al., 2001; Rutter et al., 1970). The child's mother rated the 26 items on a 3-point scale (0 = *does not apply*, 1 = *applies somewhat*, 2 = *certainly applies*) that related to childhood symptoms of anxiety, depression (e.g., "often worries, worried about many things"), and behavioral conduct disorders (e.g., "bullies other children"). Higher scores indicated worse behavior associated with poorer psychological health.

Second, psychological well-being was assessed via the Harter global self-worth subscale. Children's self-perception of psychological well-being was indexed using the global self-worth subscale of the perceived competence

Table 1. Independent Variables.

Predictor	Equation/measure ^a	Tools/example items	Scoring/notes
<i>Interior density</i>			
Home	No. of persons per room	Calculated from observations and interview responses	Excluded halls/attics/closets/garages
FCB	No. of persons sharing FCB	—	—
<i>Children's perceived crowding</i>			
Home	9 items, 3-point scale (<i>always, sometimes, never</i>)	See Online Appendix A for children's home (9) and FCB (9) perceived crowding items	Coded items were summed to yield total house and FCB crowding scores
FCB	The nine 3-point scale items used to assess home crowding items were modified to focus on FCB	—	—
<i>Design attribute</i>			
FCB window area ^b	Height $\times$ width (in ²)	Tape measure	Summed bedroom window areas
FCB ceiling height ^c	Lowest point (ft)	Tape measure	Only lowest point used in analysis
FCB volume (ft ³) ^d	Length $\times$ width $\times$ average ceiling height	Tape measure	Accounted for three dimensions
FCB depth	FCB depth was calculated as indicated in the "Introduction" section based on sketched home floor plans	—	Depth diagrams were created based on sketched home floor plans
Mean home depth	Depth of all spaces was calculated as indicated in the "Introduction" section; mean home depth was the sum of all individual space depths/($K - 1$), where K = No. of spaces	—	Depth diagrams were created based on sketched home floor plans
Home permeability (see the "Introduction" section for a description)	$RRA = RA/D_K$, where $RA = 2(\text{Mean depth} - 1)/(K - 2)$ $D_K = RA$ for diamond-shaped complexes of K cells $= 2[K(\log 2((K + 2)/3) - 1) + 1]/[(K - 1)(K - 2)]$ and K = number of spaces	—	Calculated as indicated in the "Introduction" section, based on sketched home floor plans
<i>Covariates</i>			
Home type	Recorded during observations	Detached, single-family home without subunits; mobile home or trailer; duplex/row house with ≤ 4 units; or multifamily building with > 4 units (e.g., apartment building)	—
Home clutter	4 items; 3-point scale (<i>little, some, chaos</i>). See Online Appendix B for operational definitions and photos	How much clutter is there in the room (bedroom average, bathroom average, kitchen, living area)?	Averaged coded items to yield a mean home clutter score
FCB clutter	1 item, 3-point scale (<i>little, some, chaos</i>). See Online Appendix B for operational definitions and photos	How much clutter is there in the FCB?	—
Gender	Recorded during interview	—	—

(continued)

Table 1. (continued)

Predictor	Equation/measure ^a	Tools/example items	Scoring/notes
Income-to-needs ratio ^c	= family cash income level/ official poverty line for family size at time	Family income and size provided by mother/ guardian	An income-to-needs ratio of 1 indicates a family at the poverty line
Age	Birth date was recorded during interview	—	Age was calculated in years

Note. FCB = focus child's bedroom; RRA = real relative asymmetry; RA = relative asymmetry.

^aMeasures recorded by trained research assistants during the home interview visit, unless otherwise noted.

^bWall area was not recorded; therefore, the window-to-wall area ratio could not be calculated; window treatment presence and use were also not recorded.

^cBoth the highest and lowest points of FCB ceilings were recorded, but only the lowest point was used in analyses.

^dVolume captured both two- and three-dimensional attributes of the FCB size that could affect perceptions of interior density.

^eThe income-to-needs ratio is a per capita index of household income.

scale (Cronbach's $\alpha = .67$; Harter, 1982). This six-item subscale required that children complete forced-choice format items, selecting which of two opposite behavioral descriptions were "really true or sort of true of you." Sample items included the following: "Some kids like the kind of person they are" versus "other kids often wish they were someone else," and "Some kids are often unhappy with themselves" versus "other kids are pretty pleased with themselves." Higher scores indicate higher self-ratings of self-worth.

Third, learned helplessness, related to human motivation (Evans, 2001b), was measured using an age-appropriate behavioral index that assessed persistence on a challenging task. A standard behavioral protocol (Glass & Singer, 1972) was adopted for 9-year-olds (Cohen, Evans, Stokols, & Krantz, 1986). The challenging task consisted of giving the focus children a puzzle in which they had to "visit" each object (e.g., an animal) on the paper by tracing over interconnecting lines without lifting the pencil or doubling back on any line. Participants were given instructions, a practice round using an easy version of the task, and then told to work on the challenging version until solved or until they felt unable to do so. For ethical reasons, children were then given a readily solvable version of the puzzle to ensure that all children solved a puzzle. The index of helplessness was persistence (time in seconds) on the first challenging puzzle. Longer length of time indicated longer persistence associated with higher task motivation. Variations of this helplessness index have been used in numerous laboratory and field studies and are sensitive to both manipulations of stimulus controllability and degree of exposure to chronic, uncontrollable stressors (Cohen, 1980; Evans & Stecker, 2004; Peterson, Maier, & Seligman, 1995).

Chronic physiological stress outcomes. High interior density and other environmental stressors, such as noise, elevate physiological stress in children (Evans, 2001a). Chronic physiological stress data were obtained using singular indices, as well as a composite index called allostatic load. Higher values indicate increased physiological stress for all six of the following variables.

Blood pressure tracks chronic stress (Krantz & Falconer, 1995). Children's resting systolic and diastolic blood pressure was recorded by trained researchers using an automated monitor (Critikon Corp., Tampa, FL: Dinamap Model 1846SXP) as the child sat quietly and read for 25 min. After discarding the initial measurement, six additional measurements were taken and averaged to achieve maximum reliability (Kamarck et al., 1992). Children were restricted from engaging in vigorous physical activity for 1 hr before these assessments.

Neuroendocrine measures (cortisol, epinephrine, and norepinephrine) were calculated from 12-hr overnight (8:00 p.m.-8:00 a.m.) total urine samples, kept on ice with a preservative (metabisulfite), and picked up the morning following the at-home interview. Four 10-mL samples were randomly extracted, processed (samples for catecholamine analysis are acidified), and deep frozen at -80°C for subsequent biochemical assays. Epinephrine and norepinephrine were assayed with high-pressure liquid chromatography with electrochemical detection (Riggin & Kissinger, 1977) and free cortisol with a radioimmune assay (Contreras, Hane, & Tyrrell, 1986). Creatine was assayed to provide a statistical control for differences in body mass and incomplete urine voiding (Baum & Grunberg, 1995; Grunberg & Singer, 1990; Tietz, 1976).

Allostatic load or "allostasis" refers to a dynamic, interactive set of physiological systems of equilibrium maintenance with which the human body continuously adjusts its normal operating range in response to physical and social demands (Evans, 2003b; Ganzel, Morris, & Wethington, 2010; McEwen, 1998; McEwen & Stellar, 1993; Seeman & McEwen, 1996; Sterling & Eyer, 1988). The ongoing maintenance of internal equilibrium increases allostatic load, which reflects chronic wear and tear on the body caused by the mobilization of resources to meet environmental demands (McEwen, 1998). An allostatic load index was created by summing dichotomized (e.g., 1 = within the top quartile of risk, 0 = not within top quartile) resting blood pressure (diastolic and systolic); overnight urinary neuroendocrine measures of cortisol, epinephrine, and norepinephrine; and BMI (Evans, 2003b; Kubzansky, Kawachi, & Sparrow, 1999; McEwen, 1998; Seeman, McEwen, Rowe, & Singer, 2001; Seeman, Singer, Rowe, Horwitz, & McEwen, 1997; Seeman, Singer, Ryff, Love, & Levy-Storms, 2002; Singer & Ryff, 1999). BMI, an index of metabolic function, reflects body fat and was calculated by dividing weight (kg) by squared height (m^2). Evidence

indicates that allostatic load is a better index of chronic stress than any singular component that constitutes this index (McEwen, 1998; Seeman et al., 2001; Seeman et al., 1997).

Procedure

Interior density, perceived crowding, ceiling height, volume, floor plan sketches, covariates, psychological health, and physiological stress measures extracted from the original longitudinal data set were initially collected during in-person interviews and observations conducted by pairs of trained research assistants (Evans, 2003b). Participants were informed that the study's purpose was to explore housing and child well-being. Floor plan sketches were used to calculate depth and permeability, according to Hillier and Hanson's (1984) procedures, by trained research assistants who had no knowledge of participants' residence or outcome measures.

Data analysis. IBM SPSS Statistics for Windows (IBM Corp., Version 24) was used for statistical analyses. Pearson's product-moment correlation coefficients were first calculated for FCB and home interior density, perceived FCB and home crowding, FCB and home clutter, and home type (Aim 1). Regression models also examined interior density as a predictor of children's perceived home and bedroom crowding, accounting for home type, clutter, income-to-needs, gender, and age (Aim 1). Then, regression models (Aim 2) investigated the potential interactive effects of perceived crowding (home and FCB) and each design attribute (home depth, home permeability, FCB depth, FCB ceiling height, FCB window area, FCB volume) on the dependent variables. All regression models included home type, clutter, income-to-needs, gender, and age. For comparison, regressions were also run with home and FCB density—not perceived crowding—measures.

As little prior work has examined the potential for design attributes to moderate effects of crowding on well-being, two measures were taken to address the exploratory analytical approach to Aim 2. First, the error from multiple significance testing, or running of many (54 total) statistical tests, was corrected using a method called false discovery rate (FDR; Benjamini & Hochberg, 1995; Benjamini & Yekutieli, 2001, 2005; Genovese, Lazar, & Nichols, 2002; Yekutieli & Benjamini, 1999). Unlike Bonferroni corrections that focus on controlling the probability of obtaining at least one false positive, FDR corrections have more power and control the proportion of falsely rejected hypotheses within significant results. FDR procedures adjust original p values and are reported as q values (Storey, 2003). In this study, the q -value cutoff was set to .1, meaning that the probability of obtaining a false

positive was 10% and that 90% of the significant results were expected to be true positive. A q value of .1, and not .05, was selected due to the relatively small sample size and preliminary nature of this study; results below the .1 level suggest findings worthy of further investigation (Dahiru, 2008; Fisher, 1925, 1935; Gigerenzer, Krauss, & Vitouch, 2004). The R software (version 3.0.1) package “qvalues” was used to calculate corresponding q values (Storey, Bass, Dabney, & Robinson, 2015) for two sets of Aim 2 analyses. Second, two sets of analyses were run based on whether hypotheses related to specific design attributes were a priori (based on previous literature) or exploratory (little or no previous literature). One-sided tests were run for the 27 a priori hypotheses that included FCB ceiling height, window area, and mean home depth as design attribute moderators. Resulting FDR q values were divided by 2 to reflect the one-sided tests. The remaining 27 exploratory tests examined FCB depth, FCB volume, and home permeability via two-tailed tests.

Results

Participant sociodemographics and descriptive information about their home environments are displayed in Table 2. Forty-two percent of participating children shared a bedroom with at least one other person. Parents reported that all participating children had a place to be alone, 81% of which were separate, closed spaces that did not open onto other high activity spaces such as kitchens or living areas. Only 19% of participants lived in multifamily units and 14% in trailers. Fifty percent of families resided in detached, single-family homes and another 17% in duplexes or row houses. With the exception of one four-story unit, all multifamily units were three stories or less.

Aim 1: Density and Perceived Crowding

Interior density and perceived crowding measures were significantly correlated as expected, although modestly (Table 3). Negative correlation coefficients indicated that increases in interior density were, as expected, associated with more perceived crowding (lower scores). Correlation coefficients were nearly equal for comparisons between both FCB interior density and perceived crowding ($r = -.34$) and home interior density and perceived crowding ($r = -.35$, respectively). Home interior density was also significantly correlated with perceived FCB crowding ($r = -.27$). FCB and home interior density, as well as perceived crowding measures, were also significantly positively correlated ($r = .57$ and $r = .55$, respectively). Clutter and perceived crowding measures (FCB: $r = -.16$; Home: $r = -.25$), along

Table 2. Participant Sociodemographic and Descriptive Statistics.

Variable	n	Total			Variable	n	M	SD	Range
		Variable levels	No.	%					
Gender	181	Male	90	50.0	Age (years)	181	9	1.25	6-12
		Female	91	50.0	Income-to-needs ratio ^a	181	1.14	0.65	0.10-3.74
Ethnicity	181	White	159	87.8	Home clutter (0 = little to 2 = chaos)	181	0.61	0.50	0-2
		Black	10	5.5					
		Asian	1	0.6					
		LatinX	1	0.6	FCB clutter (0 = little to 2 = chaos)	181	0.87	0.70	0-2
		Other	10	5.5	Home area (SF)	90	1,007.85	371.79	131.67-2,366.24
Home ownership	181	Own	57	32.0	No. of rooms in home/unit	181	10.47	2.80	5-19
		Rent	124	68.0	Home interior density (persons/room)	181	0.66	0.21	0.25-1.25
Home type ^b	180	Detached, single-family	90 (12/77)	50.0	FCB density (persons/FCB)	180	1.49	0.66	1-4
		Mobile home/trailer	25 (6/19)	13.9					
		Duplex/row house	30 (4/26)	16.7	Home crowding score (0 = high to 18 = low)	180	11.80	3.46	2-18
		Multifamily building	35 (8/25)	19.4	FCB crowding score (0 = high to 18 = low)	164	11.64	4.21	0-18
No. of families in home (per 1 unit)	178	1 family	30	17.4	FCB window area (in ²)	88	1,503.09	823.37	442-5,427
		More than 1 family	147	82.6	FCB ceiling height, high (ft)	132	7.61	0.90	6.33-14.31
					FCB ceiling height, low (ft)	132	7.07	1.64	2.17-14.31
Shared FCB	180	No	105	58.3	FCB volume (ft ³)	89	868.53	282.78	440.77-1,846.57
		Yes	75	41.7	FCB depth	181	27.97	14.39	7-79
Place for child to be alone ^c	177	Separate, closed	144	81.4	Mean home depth	181	2.56	0.51	1.60-3.85
		Separate, open	8	4.5					
		Alcove in a room	25	14.1					
		No space alone	0	0	Home permeability	181	0.68	0.15	0.37-1.09

Note. FCB = focus child's bedroom; SF = square footage.

^aRefer to Table 1 for a definition.

^bThe first number in parentheses indicates the number of participants with one single family in their residential unit. The second number indicates the number of participants living with more than one family in their residential unit, including the addition of a single grandparent.

^c"Closed" and "open" indicate whether the separate room does or does not, respectively, open onto a living room, kitchen, or high activity center.

Table 3. Pearson Correlations Among Perceived Crowding and Interior Density Measures.

Variable	1	2	3	4	5	6	7
1. FCB interior density	—	-.34**	.04	.57**	-.38**	.04	-.03
2. FCB crowding		—	-.16*	-.27**	.55**	-.19*	-.01
3. FCB Clutter			—	.18*	-.10	.61**	.02
4. Home interior density				—	-.35**	.31**	-.003
5. Home crowding					—	-.25**	-.16*
6. Home clutter						—	.16*
7. Home type							—

Note. FCB = focus child's bedroom.

*Significant ($p < .05$). **Significant ($p < .001$).

Table 4. Regression of Children's Perceived Home and FCB Crowding on Home and FCB Interior Density, Accounting for Home Type, Clutter, Income-to-Needs, Gender, and Age.

Outcome	Total R^2	ΔR^2	F	df	Predictor	β (SE)	p
Home crowding	.17	.05	5.92**	6,172	Home interior density	-4.80 (1.25)	<.001**
					Home clutter	-0.50 (0.31)	.106
					Home type	-0.34 (0.21)	.101
					Income-to-needs	0.39 (0.40)	.333
					Gender	0.29 (0.49)	.550
					Age	0.19 (0.19)	.339
FCB crowding	.16	.04	4.98**	6,155	FCB interior density	-2.34 (0.47)	<.001**
					FCB clutter	-0.92 (0.50)	.042*
					Home type	-0.10 (0.27)	.713
					Income-to-needs	0.78 (0.49)	.111
					Gender	-0.44 (0.63)	.489
					Age	0.22 (0.25)	.386

Note. FCB = focus child's bedroom.

*Significant ($p < .05$). **Significant ($p < .001$).

with home type and perceived home crowding ($r = -.16$), were also modestly and significantly correlated. Regression analyses further indicated that home and FCB interior density significantly predicted children's perceived home and FCB crowding ($p < .001$, Table 4). Home type, home clutter,

income, gender, and age were not significant predictors in any Aim 1 model; FCB clutter, however, was a significant predictor of perceived FCB crowding ($p < .05$, Table 4).

Aim 2: Child Outcomes of Crowding

Table 5 and Figure 3a-e display the five *a priori* regression models (out of 27) that contained significant interaction terms after FDR corrections for multiple testing. Table 5 reports results of significant interactions between perceived home crowding and FCB ceiling height on physiological health outcomes. Figure 3a-e illustrates the direction of these interactions, with higher values indicating poorer outcomes. Continuous perceived crowding and FCB ceiling height variables were dichotomized for illustrative purposes only (Figure 3). All regression analyses maintained the original continuous variables. Results indicated a significant main effect of FCB ceiling height such that lower FCB ceiling heights were associated with poorer participant physiological health outcomes, overall, when compared with outcomes of participants with higher FCB ceiling heights. Results also suggested that FCB ceiling height buffered the negative effects of perceived home crowding on physiological stress outcomes, with the exception of cortisol, as anticipated. Higher FCB ceiling heights were associated with significantly lower resting systolic ($\beta = 0.80$, $q < .05$; Figure 3a) and diastolic blood pressure ($\beta = 0.61$, $q < .05$; Figure 3b) and norepinephrine ($\beta = 0.02$, $q < .05$; Figure 3d), as well as marginally significantly lower epinephrine ($\beta = 0.02$, $q < .10$; Figure 3c) and allostatic load ($\beta = 0.11$, $q < .10$; Figure 3e), among participants who reported more perceived home crowding when compared with those who reported less. In other words, higher ceiling heights were associated with better physiological health outcomes overall and greater attenuated linkages between perceived home crowding and physiological health outcomes among participants who reported more perceived home crowding when compared with those who reported less perceived home crowding.

No significant interactions were found between perceived home crowding and window area or mean home depth on physiological health outcomes. Home type, home clutter, income, and gender were also not significant predictors in Aim 2 models; age was significantly associated with perceived home crowding in epinephrine and norepinephrine models ($p < .05$). In addition, no significant interactions were found between FCB ceiling height, window area, or mean home depth and perceived home crowding on psychological health outcomes (psychological health [Rutter], psychological well-being [Harter global self-worth],¹ and learned helplessness). Furthermore, none of the 27 exploratory models

Table 5. A Priori Regression Models of Children's Physiological Stress on Perceived Home Crowding by Design Attribute Interaction, Accounting for Home Type, Clutter, Income-to-Needs, Gender, and Age.

A priori models	Outcome ^a	Moderator (Perceived Home Crowding x Design)					p	q/2
		R ²	ΔR ²	F	df	β (SE)		
	BP (resting systolic)	.204	.15	2.687*	8,84	0.798 (0.284)	.006*	.032 [^]
	BP (resting diastolic)	.167	.13	2.098*	8,84	0.610 (0.204)	.004*	.032 [^]
	Epinephrine (log)	.164	.12	2.128*	8,87	0.018 (0.008)	.031*	.084 [^]
	Norepinephrine (log)	.256	.21	3.737*	8,87	0.015 (0.005)	.007*	.032 [^]
	Allostatic load	.207	.15	2.480*	8,76	0.107 (0.045)	.021*	.072 [^]

Note. FCB = focus child's bedroom; BP = blood pressure.

^aPsychological health outcomes included psychological health (Rutter), psychological well-being (Harter), and learned helplessness (no significant results were found); chronic physiological stress outcomes included resting systolic and diastolic BP, neuroendocrine measures (cortisol, epinephrine, and norepinephrine), and allostatic load.

^bMain effect of design predictor was also significant ($p < .05$).

^cMain effect of perceived home crowding was also significant ($p < .05$).

[^] $p < .05$ (significant). ^q $q < .10$ (significant).

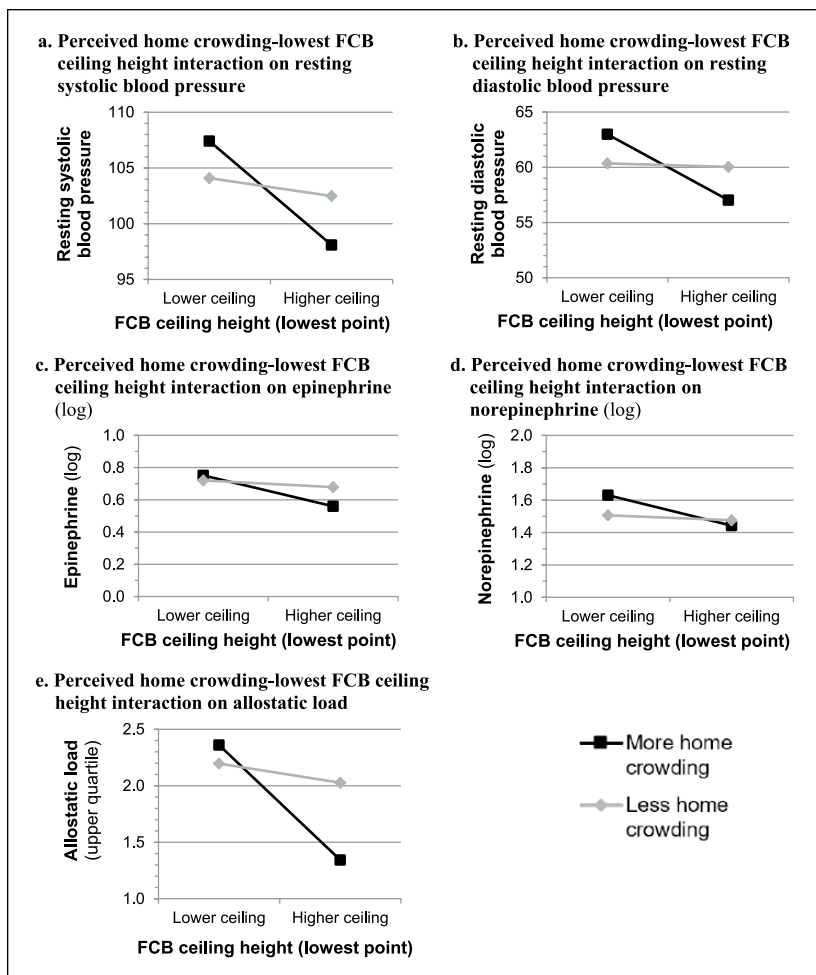


Figure 3. Significant interactions between perceived home crowding and FCB ceiling height on psychological health and physiological stress outcomes after accounting for home type, clutter, income-to-needs, gender, and age. *Note.* Perceived crowding and FCB ceiling height variables were dichotomized (median split) in interaction graphs for descriptive purposes only. Regression analyses of continuous variables were conducted. Higher outcome values in this figure are associated with poorer outcomes. FCB = focus child's bedroom.

examining interactions between FCB depth, FCB volume, or home permeability and perceived home crowding on psychological health and physiological stress outcomes yielded significant results ($q > .10$).

Remaining analyses of interactions between perceived FCB crowding and design attributes on psychological health and physiological stress also revealed no significant results for either the *a priori* or exploratory hypotheses ($q > .10$). Similarly, no significant results were found when analyzing interactions between residential interior density (home and FCB)—rather than perceived crowding—and design attributes on psychological health or physiological stress outcomes. Interaction terms were not significant in the 27 *a priori* or 27 exploratory models ($q > .10$).

Discussion

The purpose of this study was to (a) examine the relation between residential interior density and children's perceived crowding and (b) investigate the potential of design attributes to buffer the adverse effects of perceived crowding on children's psychological health and physiological stress. Aim 1 results yielded consistent support for the hypothesis that higher residential interior density is associated with greater perceived crowding both in the overall home and for children's bedrooms. Given that one of the primary contributors to perceived crowding is difficulties in regulating social interaction, the significant associations between bedroom and household interior density and perceived crowding make sense. Homes and bedrooms with higher interior densities may make it more difficult for children to be away from people when desired, and therefore they perceive crowding related to high density. Although not a surprising finding, it is also worth noting that the child's bedroom interior density was significantly correlated with perceptions of not only perceived bedroom crowding but also household crowding (Aim 1, Table 3). The same pattern was uncovered for home interior density and perceived bedroom and household crowding. These results suggest that interior density of a child's bedroom or home may be associated with perceived crowding, and that both bedroom and home design attributes could potentially serve as moderators of the negative effects of crowding.

Five *a priori* Aim 2 models indicated that higher FCB ceiling heights were associated with attenuated linkages between perceived home crowding and several markers of chronic physiological stress, especially among more crowded participants. Prior work examining effects of ceiling height on adults' personal space perceptions offers a possible explanation for the potential moderating effect of ceiling height found in the present study. Adults in rooms with a higher ceiling perceived having more personal space when compared with adults in rooms of the same size with a lower ceiling height (Savinar, 1975). In the present study, child participants in bedrooms with higher versus lower ceilings may have, similarly, perceived greater amounts of personal space and thus experienced attenuated effects of perceived crowding. Another possible

explanation is that higher ceilings increased perceived volume or overall room size, in turn reducing perceived crowding. Higher ceilings could also be associated with bedroom space use, such as the inclusion of bunkbeds that increase available floor space and lower perceptions of crowding. Future work can pursue each of these possible models for why ceiling height might affect perceived crowding. The first explanation can be explored by assessing the size and shape of children's personal space bubbles; the second can be explored via a standardized psychophysical technique, such as magnitude estimation, to document perceptions of room size. Study results also expanded upon findings that suggest children living in high interior density homes who can retreat to a space of their own, such as a bedroom, suffer fewer negative effects of high interior density (Wachs & Gruen, 1982). Children who perceive having more control over social interactions and the ability to control the number of people in their space by retreating to a bedroom may have experienced attenuated effects of high interior density and perceived crowding in this study (Altman, 1975; Evans et al., 2001; Rodin, 1976). Design attributes of the bedroom, including ceiling height based on study results, may also play a role.

The Aim 2 methodological approach consisted of running 54 statistical tests and yielded only five significant results after FDR corrections, so results of this exploratory work should be interpreted with caution. However, ceiling height moderated effects of perceived crowding on five of the six physiological stress outcomes examined, as expected, and 90% or 4.5 of the five significant results are expected to be "true positive" based on FDR corrections. More generally, exploratory analysis is a critical first step in the early stages of inquiry, particularly for questions that have seen little or no prior analysis (Winkel, Saegert, & Evans, 2009). Results offer some insights to guide future research on the potential for design attributes to attenuate negative effects of perceived crowding on children's well-being.

The lack of significant results on psychological well-being may be due, in part, to greater sensitivity of physiological markers than self-reported measures. The divergence of findings for physiological and self-reported measures also serves as an important reminder that overreliance on self-reported measures is a major impediment to scientific progress in environmental psychology (Winkel et al., 2009). Furthermore, the lack of significant exploratory model results and significant moderation by other design attributes (home permeability, mean home depth, and FCB window area, volume, and depth) may have been partially due to the limited variability among design attributes and a small sample size that precluded use of advanced statistical models that simultaneously examine all design attributes. Additional work is needed to explore how these design attributes contribute to perceived crowding and whether they affect psychological outcomes.

The study contributes to the crowding literature in four ways. First, few studies have examined the potential of design attributes to moderate the relations between perceived crowding and psychological well-being or with physiological stress generally, and this is especially true for children. The relation between interior density and perceived crowding may differ among children when compared with adults. Residential interior density alone may not be enough to induce adverse effects of perceived crowding among child participants, as suggested by the modest correlations found in this study. Other social stressors within and physical characteristics of the home (e.g., parenting, acoustics, window height, window treatments, housing quality) may influence the interior density–perceived crowding relation (Schiffenbauer et al., 1977; Stokols, 1972b). Furthermore, social interaction and privacy may be regulated through both psychological and environmental coping processes (Altman, 1975; Archea, 1977; Evans et al., 1996). Second, standardized indices of children's psychological health and physiological stress were used. Most studies on crowding (via multiple measures) and children have not relied on standardized instruments to assess mental health, and physiological stress outcomes are rare. Third, this study is one of few that explored the predictive validity of depth and permeability, two space syntax theoretical constructs.

Finally, this study provided evidence that FCB ceiling height has the potential to buffer adverse effects of perceived home crowding, especially among children reporting higher levels of perceived home crowding, and positively affect children's health and well-being. The potential for ceiling height to attenuate the negative effects of perceived crowding is an especially critical finding considering current and anticipated increases in residential interior density, especially among low-income populations and in urban areas where square footage is limited. Additional studies must explore whether the potential well-being benefits of raising ceiling heights in high-density living conditions outweigh increases in construction costs.

Limitations

Like much of the crowding literature, this study was cross-sectional and concerns about causality are tantamount. Although models were adjusted for income—and somewhat addressed self-selection as parents, not children, select where they live—there are many plausible alternative explanations that might explain the results of the study. Length of exposure to high interior density in the home and bedroom, previous living conditions, parent behavior and well-being, noise, and other stressors in and outside of the home were also not considered in this study. At low levels of residential

interior density, the relation between interior density, perceived crowding, and negative outcomes likely depends upon other psychological, environmental, and situational factors (Gillis, 1979). At higher levels of interior density, the probability of experiencing adverse effects of perceived crowding increases, as well as other factors associated with high interior density influencing discomfort.

Second, the association between housing design elements and children's well-being may vary by individual child characteristics (e.g., age) and over time. For instance, the interactive effects of ceiling height and perceived crowding may change over time as children age. Associations between perceived crowding and well-being may be stronger for younger children who spend more time inside the home when compared with adolescents (Leventhal & Newman, 2010). Conversely, children's growth in size as they age may make a bedroom feel increasingly smaller. As children enter into adolescence, however, they may be able to more easily escape the pressures of unwanted social interaction by leaving the residence. A future longitudinal or intervention study would help establish causality (Evans et al., 1996). Third, the small sample size and low variability among study variables may also partially explain the lack of significant results for design attributes other than ceiling height. A wider and more representative range of residential interior densities, housing design attributes (window area, ceiling height), and child sociodemographic factors, in addition to a larger sample size, would strengthen the study design and generalizability. A larger sample size would also allow use of more advanced statistical models (e.g., structural equation modeling) to simultaneously examine all design attributes and outcomes. Multiple design attributes likely affect how density is perceived, so a simultaneous examination of multiple design attributes—rather than one-at-a-time—is needed. The reliability of perceived crowding measures used with young children also requires further testing. Additional interior design attributes such as openness, visual access, acoustic treatments, and lighting may also affect perceptions of residential interior density. Furthermore, study results were inconsistent with prior work on depth. Mean home depth did not moderate the interior density–psychological distress relation among children as in prior studies of college students in off-campus housing (Evans et al., 1996). Per parent reports, the child participants all had a place to be alone and resided in low-rise buildings (4 stories or less) in rural areas and small towns. Effects of design attributes, including depth, in taller buildings and urban areas may differ. Additional work is needed to examine the effects of depth and other design attributes on the perceived crowding–psychological health relationship among young children.

Implications and Conclusions

Crowding is a widely acknowledged source of adult stress that detracts from housing quality, yet little is known about the effects of crowding on children. Study results suggest associations between interior density and perceived crowding among children, and that ceiling height might moderate negative effects of perceived crowding on children's well-being. High interior density conditions negatively affect children's well-being, including academic achievement, behavior, health, SES, and adult well-being (Lopoo & London, 2016; Solari & Mare, 2012). Low-income populations facing multiple environmental and psychological stressors are, furthermore, most likely to be affected by high interior density and perceived crowding (Evans & English, 2002; Gillis, 1979; Saegert, 1982). Identifying design elements that enable people to cope better with high interior densities, as well as other suboptimal living conditions, is necessary to guide housing policy and accommodate expected increases in high interior density living conditions (Mackun, Wilson, Fischetti, & Goworowska, 2011). This knowledge can inform the design, construction, and renovation of existing and future high interior density housing developments without introducing additional or worsening existing housing stressors.

This study explored the potential of interior design attributes to modify children's responses to perceived crowding and offers directions for future research. The study examined design as a potential moderator of perceived crowding, an environmental phenomenon with well-documented effects on human behavior. Results suggest that other physical design features, beyond available floor space, may have some utility in attenuating the negative effects of high interior density associated with crowding, typically measured by a lack of space or interference with desired levels of social interaction. These additional design features may provide designers, who often make claims about how design can affect perceptions of available space, with a richer evidence-based toolbox from which to address high interior density.

Increasing residential ceiling heights or adjusting other design attributes that affect perceptions of ceiling height, especially when increasing square footage is not possible, could buffer adverse effects of perceived crowding on residents, including children.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was supported by funding from the John D. and Catherine T. MacArthur Foundation: How Housing Matters to Families & Communities; the John D. and Catherine T. MacArthur Foundation Network on Socioeconomic Status and Health; and the William T. Grant Foundation.

Note

1. Earlier analyses that excluded home type and clutter covariates revealed an additional significant result. Focus child's bedroom (FCB) volume moderated the effects of home crowding on perceived psychological well-being (Harter), but not in the anticipated direction. Increases in FCB volume were associated with a slight increase in Harter global self-worth subscale scores for more crowded participants compared with a slight decrease in scores for less crowded participants. Less crowded participants experienced better perceived psychological well-being overall when compared with more crowded participants, but rated well-being slightly lower in higher ceiling height conditions, contrary to hypotheses.

Supplemental Material

Supplemental material for this article is available online.

ORCID iD

Kimberly A. Rollings  <https://orcid.org/0000-0002-3091-3340>

References

- Aiello, J. R., Epstein, Y. M., & Karlin, R. A. (1974, April). *Methodological and conceptual issues in crowding*. Paper presented at the Western Psychological Association, San Francisco, CA.
- Aiello, J. R., Nicosia, G., & Thompson, D. E. (1979). Physiological, social, and behavioral consequences of crowding on children and adolescents. *Child Development*, 50, 195-202. doi:10.1111/j.1467-8624.1979.tb02994.x
- Altman, I. (1975). *The environment and social behavior*. Monterey, CA: Brooks-Cole.
- Archea, J. (1977). The place of architectural factors in behavioral theories of privacy. *Journal of Social Issues*, 33, 116-137. doi:10.1111/j.1540-4560.1977.tb01886.x
- Baum, A., & Davis, G. (1976). Spatial and social aspects of crowding perception. *Environment & Behavior*, 8, 527-544. doi:10.1177/001391657684003
- Baum, A., & Davis, G. (1980). Reducing the stress of high-density living: An architectural intervention. *Journal of Personality and Social Psychology*, 38, 471-481. doi:10.1037/0022-3514.38.3.471
- Baum, A., Gatchel, R., Aiello, J. R., & Thompson, D. (1981). Cognitive mediation of environmental stress. In J. H. Harvey (Ed.), *Cognition, social behavior, and the environment* (pp. 513-533). Hillsdale, NJ: Lawrence Erlbaum.

- Baum, A., & Grunberg, N. (1995). Measurement of stress hormones. In S. Cohen, R. C. Kessler, & L. Gordon (Eds.), *Measuring stress* (pp. 175-192). New York, NY: Oxford University Press.
- Baum, A., & Paulus, P. (1987). Crowding. In D. Stokols & I. Altman (Eds.), *Handbook of environmental psychology* (pp. 533-570). New York, NY: John Wiley.
- Baum, A., & Valins, S. (1979). Architectural mediation of residential density and control: Crowding and the regulation of social contact. *Advances in Experimental Social Psychology*, 12, 131-175. doi:10.1016/s0065-2601(08)60261-0
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57, 289-300.
- Benjamini, Y., & Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29, 1165-1188.
- Benjamini, Y., & Yekutieli, D. (2005). Quantitative trait loci analysis using the false discovery rate. *Genetics*, 171, 783-790. doi:10.1534/genetics.104.036699
- Booth, A., & Edwards, J. (1976). Crowding and family relations. *American Sociological Review*, 41, 289-308. doi:10.2307/2094476
- Boyle, M., & Jones, S. J. (1985). Selecting measures of emotional and behavioral disorders of children for use in general populations. *Journal of Clinical Psychology and Psychiatry*, 26, 137-159. doi:10.1111/j.1469-7610.1985.tb01634.x
- Bradley, R. H., & Caldwell, M. (1984). The home inventory and family demographics. *Developmental Psychology*, 20, 315-320. doi:10.1037/0012-1649.20.2.315
- Butler, D. L., & Steuerwald, B. L. (1991). Effects of view and room size on window size preferences made in models. *Environment & Behavior*, 23, 334-358. doi:10.1177/0013916591233005
- Cleland, V., Timperio, A., Salmon, J., Hume, C., Baur, L. A., & Crawford, D. (2010). Predictors of time spent outdoors among children: 5-year longitudinal findings. *Journal of Epidemiology & Community Health*, 64, 400-406. doi:10.1136/jech.2009.087460
- Cohen, S. (1980). Aftereffects of stress on human performance and social behavior: A review of research and theory. *Psychological Bulletin*, 88, 82-108.
- Cohen, S., Evans, G. W., Stokols, D., & Krantz, D. S. (1986). *Behavior, health, and environmental stress*. New York, NY: Plenum.
- Collins, B. L. (1975). *Windows and people: A literature survey*. Washington, DC: Department of Commerce/National Bureau of Standards.
- Contreras, L. N., Hane, S., & Tyrrell, J. B. (1986). Urinary cortisol in the assessment of pituitary adrenal function: Utility of 24-hour and spot determinations. *Journal of Clinical Endocrinology & Metabolism*, 62, 965-969. doi:10.1210/jcem-62-5-965
- Cox, V. C., Paulus, P., McCain, G., & Karlovac, M. (1982). The relationship between crowding and health. In A. Baum & J. E. Singer (Eds.), *Advances in environmental psychology* (Vol. 4, pp. 271-294). Hillsdale, NJ: Lawrence Erlbaum.
- Dahiru, T. (2008). P-value, a true test of statistical significance? A cautionary note. *Annals of Ibadan Postgraduate Medicine*, 6, 21-26. doi:10.4314/aipm.v6i1.64038

- Evans, G. W. (1979). Behavioral and physiological consequences of crowding in humans. *Journal of Applied Social Psychology*, 9, 27-46. doi:10.1111/j.1559-1816.1979.tb00793.x
- Evans, G. W. (2001a). Environmental stress and health. In A. Baum, T. Revenson, & J. E. Singer (Eds.), *Handbook of health psychology* (pp. 365-385). Mahwah, NJ: Lawrence Erlbaum.
- Evans, G. W. (2001b). Housing quality and children's socioemotional health. *Environment & Behavior*, 33, 389-399. doi:10.1177/00139160121973043
- Evans, G. W. (2003a). The built environment and mental health. *Journal of Urban Health: Bulletin of the New York Academy of Medicine*, 80, 536-555.
- Evans, G. W. (2003b). A multimethodological analysis of cumulative risk and allostatic load among rural children. *Developmental Psychology*, 39, 924-933. doi:10.1037/0012-1649.39.5.924
- Evans, G. W. (2004). The environment of childhood poverty. *American Psychologist*, 59, 77-92. doi:10.1037/0003-066x.59.2.77
- Evans, G. W. (2006). Child development and the physical environment. *Annual Review of Psychology*, 57, 423-451. doi:10.1146/annurev.psych.57.102904.190057
- Evans, G. W., & English, K. (2002). The environment of poverty: Multiple stressor exposure, psychophysiological stress, and socioemotional adjustment. *Child Development*, 73, 1238-1248. doi:10.1111/1467-8624.00469
- Evans, G. W., & Lepore, S. J. (1993). Household crowding and social support: A quasi-experimental analysis. *Journal of Personality and Social Psychology*, 65, 308-316. doi:10.1037/0022-3514.65.2.308
- Evans, G. W., Lepore, S. J., & Schroeder, A. (1996). The role of interior design elements in human responses to crowding. *Journal of Personality and Social Psychology*, 70, 41-46. doi:10.1037/0022-3514.70.1.41
- Evans, G. W., Lepore, S. J., Sejwal, B., & Palsane, M. N. (1998). Chronic residential crowding and children's well-being: An ecological perspective. *Child Development*, 69, 1514-1523. doi:10.1111/j.1467-8624.1998.tb06174.x
- Evans, G. W., Lercher, P., & Kofler, W. W. (2002). Crowding and children's mental health: The role of house type. *Journal of Environmental Psychology*, 22, 221-231. doi:10.1006/jevp.2002.0256
- Evans, G. W., Maxwell, L. E., & Hart, B. (1999). Parental language and verbal responsiveness to children in crowded homes. *Developmental Psychology*, 35, 1020-1023. doi:10.1037/0012-1649.35.4.1020
- Evans, G. W., Palsane, M. N., Lepore, S. J., & Martin, J. (1989). Residential density and psychological health: The mediating effects of social support. *Journal of Personality and Social Psychology*, 57, 994-999. doi:10.1037/0022-3514.57.6.994
- Evans, G. W., Rhee, E., Forbes, C., Allen, K. M., & Lepore, S. J. (2000). The meaning and efficacy of social withdrawal as a strategy for coping with chronic residential crowding. *Journal of Environmental Psychology*, 20, 335-342. doi:10.1006/jevp.1999.0174

- Evans, G. W., & Saegert, S. (2000). Residential crowding in the context of inner city poverty. In S. E. Wapner, J. E. Demick, T. E. Yamamoto, & H. E. Minami (Eds.), *Theoretical perspectives in environment-behavior research* (pp. 247-268). New York, NY: Plenum.
- Evans, G. W., Saegert, S., & Harris, R. (2001). Residential density and psychological health among children in low-income families. *Environment & Behavior*, 33, 165-180. doi:10.1177/00139160121972936
- Evans, G. W., & Stecker, R. (2004). Motivational consequences of environmental stress. *Journal of Environmental Psychology*, 24, 143-165. doi:10.1016/S0272-4944(03)00076-8
- Fisher, R. A. (1925). *Statistical methods for research workers*. Edinburgh, Scotland: Oliver and Boyd.
- Fisher, R. A. (1935). *The design of experiments*. Edinburgh, Scotland: Oliver and Boyd.
- Forsyth, A., Oakes, J. M., Schmitz, K. H., & Hearst, M. (2007). Does residential density increase walking and other physical activity? *Urban Studies*, 44, 679-697. doi:10.1080/00420980601184729
- Ganzel, B. L., Morris, P. A., & Wethington, E. (2010). Allostasis and the human brain: Integrating models of stress from the social and life sciences. *Psychological Review*, 117, 134-174. doi:10.1037/a0017773
- Genovese, C. R., Lazar, N. A., & Nichols, T. (2002). Thresholding of statistical maps in functional neuroimaging using the false discovery rate. *NeuroImage*, 15, 870-878. doi:10.1006/nimg.2001.1037
- Gigerenzer, G., Krauss, S., & Vitouch, O. (2004). The null ritual: What you always wanted to know about significance testing but were afraid to ask. In D. Kaplan (Ed.), *The SAGE handbook of quantitative methodology for the social sciences* (pp. 391-408). Thousand Oaks, CA: Sage.
- Gillis, A. R. (1979). Household density and human crowding: Unravelling a non-linear relationship. *Journal of Population*, 2, 104-117. doi:10.1007/bf00983015
- Glass, D. C., & Singer, J. E. (1972). *Urban stress: Experiments on noise and social stressors*. New York, NY: Academic Press.
- Grunberg, N., & Singer, J. E. (1990). Biochemical measurement. In J. Cacioppo & L. Tassinary (Eds.), *Principles of psychophysiology* (pp. 149-176). New York, NY: Cambridge University Press.
- Hanson, J. (2003). *Decoding homes and houses*. Cambridge, UK: Cambridge University Press.
- Harter, S. (1982). The perceived competence scale for children. *Child Development*, 53, 87-97. doi:10.2307/1129640
- Hillier, B., & Hanson, J. (1984). *Social logic of space*. London, England: Cambridge University Press.
- Hofferth, S. L., & Sandberg, J. F. (2004). How American children spend their time. *Journal of Marriage and Family*, 63, 295-308. doi:10.1111/j.1741-3737.2001.00295.x
- Kamarck, T., Jennings, R., Debski, T., Glicksman-Weis, E., Johnson, P., Eddy, M., & Manuck, S. B. (1992). Reliable measures of behaviorally evoked cardio-

- vascular reactivity from a PC-based test battery. *Psychophysiology*, 29, 17-28. doi:10.1111/j.1469-8986.1992.tb02006.x
- Kaya, N., & Erkip, F. (2001). Satisfaction in a dormitory building: The effects of floor height on the perception of room size and crowding. *Environment & Behavior*, 33, 35-53. doi:10.1177/00139160121972855
- Krantz, D. S., & Falconer, J. (1995). Measurement of cardiovascular responses. In S. Cohen, R. C. Kessler, & L. Gordon (Eds.), *Measuring stress* (pp. 193-212). New York, NY: Oxford University Press.
- Kubzansky, L. D., Kawachi, I., & Sparrow, D. (1999). Socioeconomic status, hostility, and risk factor clustering in the normative aging study: Any help from the concept of allostatic load? *Annals of Behavioral Medicine*, 21, 330-338. doi:10.1007/bf02895966
- Lepore, S. J., Evans, G. W., & Palsane, M. N. (1991). Social hassles and psychological health in the context of chronic crowding. *Journal of Health and Social Behavior*, 32, 357-367. doi:10.2307/2137103
- Lepore, S. J., Merritt, K., Kawasaki, N., & Mancuso, R. (1990, May). *Social withdrawal in crowded residences*. Paper presented at the Western Psychological Association, Los Angeles, CA.
- Leventhal, T., & Newman, S. (2010). Housing and child development. *Children and Youth Services Review*, 32, 1165-1174. doi:10.1016/j.childyouth.2010.03.008
- Lopoo, L. M., & London, A. S. (2016). Household crowding during childhood and long-term education outcomes. *Demography*, 53, 699-721. doi:10.1007/s13524-016-0467-9
- Mackun, P., Wilson, S., Fischetti, T., & Goworowska, J. (2011). *Population distribution and change: 2000 to 2010* (C2010BR-01). Washington, DC: U.S. Census Bureau. Retrieved from <http://www.census.gov/prod/cen2010/briefs/c2010br-01.pdf>
- Masoudinejad, S., & Hartig, T. (2018). Window view to the sky as a restorative resource for residents in densely populated cities. *Environment & Behavior*. Advance online publication. doi:10.1177/0013916518807274
- McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine*, 338, 171-179. doi:10.1056/nejm199801153380307
- McEwen, B. S., & Stellar, E. (1993). Stress and the individual: Mechanisms leading to disease. *Annals of Internal Medicine*, 119, 2093-2101. doi:10.1001/archinte.153.18.2093
- Ne'eman, E., & Hopkinson, R. G. (1970). Critical minimum acceptable window size: A study of window design and provision of a view. *Lighting Research & Technology*, 2, 17-27. doi:10.1177/14771535700020010701
- Olson, L. (1975, January). *Recommendations for environmental design*. Paper presented at the fifth annual Alpha Kappa Delta Sociological Research Symposium, Richmond, VA.
- Peterson, C., Maier, S. F., & Seligman, M. E. P. (1995). *Learned helplessness: A theory for the age of personal control*. Oxford, UK: Oxford University Press.
- Riggin, R., & Kissinger, P. (1977). Determination of catecholamines in urine by reverse-phase liquid chromatography with electrochemical detection. *Analytic Chemistry*, 49, 2109-2111. doi:10.1021/ac50021a052

- Rodin, J. (1976). Density, perceived choice, and responses to controllable and uncontrollable outcomes. *Journal of Experimental Social Psychology*, 12, 564-578. doi:10.1016/0022-1031(76)90035-4
- Rutter, M., Tizard, J., & Whitmore, K. (1970). *Education, health, and behavior*. London, England: Longman.
- Saegert, S. (1982). Environment and children's mental health: Residential density and low income children. In A. Baum & J. E. Singer (Eds.), *Handbook of psychology and health* (Vol. 2, pp. 247-271). Hillsdale, NJ: Lawrence Erlbaum.
- Savinar, J. (1975). The effect of ceiling height on personal space. *Man-Environmental Systems*, 5, 321-324.
- Schaeffer, M., Baum, A., Paulus, P., & Gaes, G. (1988). Architecturally mediated effects of social density in prison. *Environment & Behavior*, 20, 3-19. doi:10.1177/0013916588201001
- Schiffenbauer, A. I., Brown, J. E., Perry, P. L., Shulack, L. K., & Zanzola, A. M. (1977). The relationship between density and crowding: Some architectural modifiers. *Environment & Behavior*, 9(1), 3-14. doi:10.1177/001391657791001
- Seeman, T. E., & McEwen, B. S. (1996). Impact of social environmental characteristics on neuroendocrine regulation. *Psychosomatic Medicine*, 58, 459-471. doi:10.1097/00006842-199609000-00008
- Seeman, T. E., McEwen, B. S., Rowe, J. W., & Singer, B. H. (2001). Allostatic load as a marker of cumulative biological risk: MacArthur studies of successful aging. *Proceedings of the National Academy of Sciences*, 98, 4770-4775. doi:10.1073/pnas.081072698
- Seeman, T. E., Singer, B. H., Rowe, J. W., Horwitz, R., & McEwen, B. S. (1997). Price of adaptation-allostatic load and its health consequences: MacArthur studies of successful aging. *Archives of Internal Medicine*, 157, 2259-2268. doi:10.1001/archinte.157.19.2259
- Seeman, T. E., Singer, B. H., Ryff, C. D., Love, G., & Levy-Storms, L. (2002). Social relationships, gender, and allostatic load across two age cohorts. *Psychosomatic Medicine*, 64, 395-406. doi:10.1097/00006842-200205000-00004
- Singer, B. H., & Ryff, C. D. (1999). Hierarchies of life histories and associated health risks. In N. E. Adler, M. Marmot, B. S. McEwen, & J. Stewart (Eds.), *Socioeconomic status and health in industrial nations* (pp. 96-115). New York: New York Academy of Sciences.
- Solari, C. D., & Mare, R. D. (2012). Housing crowding effects on children's wellbeing. *Social Science Research*, 41, 464-476. doi:10.1016/j.ssresearch.2011.09.012
- Sterling, P., & Eyer, J. (1988). Allostasis: A new paradigm to explain arousal pathology. In S. Fisher & J. Reason (Eds.), *Handbook of life stress, cognition, and health* (pp. 629-649). New York, NY: John Wiley.
- Stokols, D. (1972a). On the distinction between density and crowding: Some implications for future research. *Psychology Review*, 79, 275-277. doi:10.1037/h0032706
- Stokols, D. (1972b). A social-psychological model of human crowding phenomena. *Journal of the American Institute of Planners*, 38(2), 72-83.

- Storey, J. D. (2003). The positive false discovery rate: A Bayesian interpretation and the q-value. *The Annals of Statistics*, 31(6), 2013-2035. doi:10.1214/aos/1074290335
- Storey, J. D., Bass, A. J., Dabney, A., & Robinson, D. (2015). Package "qvalue" (Version 2.13.0) [Package]. Retrieved from <http://github.com/jdstorey/qvalue>
- Sundstrom, E. (1975). An experimental study of crowding: Effects of room size, intrusion, self-disclosure, and self-reported stress. *Journal of Personality and Social Psychology*, 32, 645-654. doi:10.1037/0022-3514.32.4.645
- Tietz, N. (1976). *Fundamentals of clinical chemistry* (2nd ed.). Philadelphia, PA: W. B. Saunders.
- Wachs, T. D. (1989). The nature of the physical microenvironment: An expanded classification system. *Merrill Palmer Quarterly*, 35, 399-419.
- Wachs, T. D., & Gruen, G. (1982). *Early experience and human development*. New York, NY: Plenum Press.
- Wells, N. M., & Harris, J. D. (2007). Housing quality, psychological distress, and the mediating role of social withdrawal: A longitudinal study of low-income women. *Journal of Environmental Psychology*, 27, 69-78. doi:10.1016/j.jenvp.2006.11.002
- Winkel, G., Saegert, S., & Evans, G. W. (2009). An ecological perspective on theory, methods, and analysis in environmental psychology: Advances and challenges. *Journal of Environmental Psychology*, 29, 318-328. doi:10.1016/j.jenvp.2009.02.005
- Yekutieli, D., & Benjamini, Y. (1999). Resampling-based false discovery rate controlling multiple test procedures for correlated test statistics. *Journal of Statistical Planning and Inference*, 82, 171-196. doi:10.1016/S0378-3758(99)00041-5

Author Biographies

Kimberly A. Rollings is an assistant professor at the University of Notre Dame School of Architecture, with a concurrent appointment in Psychology. Her research examines effects of the built and natural environment on physical and mental health, especially among vulnerable populations.

Gary W. Evans, the Elizabeth Lee Vincent Professor of Human Ecology, Cornell University, is an environmental and developmental psychologist interested in children's environments and poverty and child development.